

Module - 5

Numerical Methods

II

Solution of linear system of equations using Gauss - Jacobi method: -

The system of linear equations, say

$$a_1x + b_1y + c_1z = d_1 \text{ ———— (1)}$$

$$a_2x + b_2y + c_2z = d_2 \text{ ———— (2)}$$

$$a_3x + b_3y + c_3z = d_3 \text{ ———— (3)}$$

will be solvable by iteration method, if

$$|a_1| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2|$$

$|c_3| > |a_3| + |b_3|$. These are known as 'conditions of convergence' by iteration methods.

On the other words, the absolute value of leading diagonal elements of the coefficient matrix should be greater than the sum of absolute values of other coefficients.

$$\text{From (1) : } x = \frac{1}{a_1} [d_1 - b_1y - c_1z]$$

$$\text{From (2) : } y = \frac{1}{b_2} [d_2 - a_2x - c_2z]$$

$$\text{From (3) : } z = \frac{1}{c_3} [d_3 - a_3x - b_3y]$$

} (A)

Take initial solution as $x^{(0)} = y^{(0)} = z^{(0)} = 0$

Then, substituting $x^{(0)} = y^{(0)} = z^{(0)} = 0$ on the RHS of (A), we get

1st approximate solution is,

$$x^{(1)} = \frac{1}{a_1} [d_1 - b_1 y^{(0)} - c_1 z^{(0)}]$$

$$y^{(1)} = \frac{1}{b_2} [d_2 - a_2 x^{(0)} - c_2 z^{(0)}]$$

$$z^{(1)} = \frac{1}{c_3} [d_3 - a_3 x^{(0)} - b_3 y^{(0)}] \quad \therefore \text{Sub. } x^{(1)}, y^{(1)}, z^{(1)} \text{ on the RHS of } \textcircled{1}, \text{ we get,}$$

2nd approximate solution is,

$$x^{(2)} = \frac{1}{a_1} [d_1 - b_1 y^{(1)} - c_1 z^{(1)}]$$

$$y^{(2)} = \frac{1}{b_2} [d_2 - a_2 x^{(1)} - c_2 z^{(1)}]$$

$$z^{(2)} = \frac{1}{c_3} [d_3 - a_3 x^{(1)} - b_3 y^{(1)}]$$

— — — — —

The iteration process may be continued till the solution set converges.

[Q] Solve by the Jacobi iteration method the system,
 $8x - 3y + 2z = 20$, $6x + 3y + 12z = 35$, $4x + 11y - z = 33$

Sol: First of all to solve the given system of equations by iteration method, we arrange the equations so that condition for convergence gets satisfied.

$$\text{i.e. } |a_1| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2| \quad \text{and} \quad |c_3| > |a_3| + |b_3|$$

Thus, writing the equations in new order, we have

$$8x - 3y + 2z = 20 \quad \text{--- ①}$$

$$4x + 11y - z = 33 \quad \text{--- ②}$$

$$6x + 3y + 12z = 35 \quad \text{--- ③}$$

so that diagonal elements are dominant in the coefficient matrix

NOTE: $|8| > |-3| + |2|$

$|11| > |4| + |-1|$

$|12| > |6| + |3|$

$$\left. \begin{array}{l} \text{From ①: } x = \frac{1}{8} (20 + 3y - 2z) \\ \text{From ②: } y = \frac{1}{11} (33 - 4x + z) \\ \text{From ③: } z = \frac{1}{12} (35 - 6x - 3y) \end{array} \right\} \text{ (A)}$$

We start with initial approximation, $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and substituting these on the RHS of (A), we get

1st approximation solution as,

$$x^{(1)} = \frac{1}{8} (20 + 3 \times 0 - 2 \times 0) = \frac{20}{8} = 2.5$$

$$y^{(1)} = \frac{1}{11} (33 - 4 \times 0 + 0) = \frac{33}{11} = 3$$

$$z^{(1)} = \frac{1}{12} (35 - 6 \times 0 - 3 \times 0) = \frac{35}{12} = 2.916667$$

Substituting $x^{(1)}$, $y^{(1)}$, $z^{(1)}$ on the RHS of (A), we get

2nd approximate solution as,

$$x^{(2)} = \frac{1}{8} (20 + 3 \times 3 - 2 \times 2.916667) = 2.8958333$$

$$y^{(2)} = \frac{1}{11} (33 - 4 \times 2.5 + 2.916667) = 2.3560606$$

$$z^{(2)} = \frac{1}{12} (35 - 6 \times 2.5 - 3 \times 2.916667) = 0.9166666$$

Substituting $x^{(2)}$, $y^{(2)}$, $z^{(2)}$ on the RHS of A, we get

3rd approximate solution as

$$x^{(3)} = 3.1543561$$

$$y^{(3)} = 2.030303$$

$$z^{(3)} = 0.8797348$$

Substituting $x^{(3)}, y^{(3)}, z^{(3)}$ on the RHS of (A), we get

4th approximate solution as

$$x^{(4)} = 3.0414299$$

$$y^{(4)} = 1.9329373$$

$$z^{(4)} = 0.8319128$$

Substituting $x^{(4)}, y^{(4)}, z^{(4)}$ on the RHS of (A), we get

5th approximate solution as

$$x^{(5)} = 3.01687$$

$$y^{(5)} = 1.969654$$

$$z^{(5)} = 0.91272$$

Substituting $x^{(5)}, y^{(5)}, z^{(5)}$ on the RHS of (A), we get

6th approximate solution as

$$x^{(6)} = 3.0104$$

$$y^{(6)} = 1.98593$$

$$z^{(6)} = 0.9582$$

Now, $x^{(7)} = 3.0158$

$$y^{(7)} = 1.9886$$

$$z^{(7)} = 0.9149$$

$$x^{(8)} = 3.0169$$

$$\text{and } y^{(8)} = 1.9865$$

$$z^{(8)} = 0.9116$$

Thus, from 7th & 8th approximations, we see that the values of x, y, z are same upto two decimal places. Stopping at this stage solution is
 $x = 3.0169,$
 $y = 1.9865$
 $z = 0.9116$ //

Q] Solve using Jacobi iteration method: Given

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

Sol:

Here, $|27| > |6| + |-1|$
 $|15| > |6| + |2|$
 $|54| > |1| + |1|$ } Thus diagonal elements are dominant in the coefficient matrix.

$$\left. \begin{array}{l} \text{From 1st eq: } x = \frac{1}{27} (85 - 6y + z) \\ \text{From 2nd eq: } y = \frac{1}{15} (72 - 6x - 2z) \\ \text{From 3rd eq: } z = \frac{1}{54} (110 - x - y) \end{array} \right\} \textcircled{A}$$

Starting with initial approximation $x^{(0)}, y^{(0)}, z^{(0)} = 0$ and substituting on the RHS of $\textcircled{A}$, we get

1st approximation as,

$$x^{(1)} = \frac{1}{27} [85 - 0 + 0] = \frac{85}{27} = 3.148$$

$$y^{(1)} = \frac{1}{15} [72 - 0 - 0] = \frac{72}{15} = 4.8$$

$$z^{(1)} = \frac{1}{54} [110 - 0 - 0] = 2.037$$

Substituting $x^{(1)}, y^{(1)}, z^{(1)}$ on the RHS of $\textcircled{A}$, we get
2nd approximation as,

$$x^{(2)} = \frac{1}{27} [85 - 6 \times 4.8 + 2 \times 2.037] = 2.157$$

$$y^{(2)} = \frac{1}{15} [72 - 6 \times 3.148 - 2 \times 2.037] = 3.269$$

$$z^{(2)} = \frac{1}{54} [110 - 3.148 - 4.8] = 1.89$$

Substituting $x^{(2)}$, $y^{(2)}$, $z^{(2)}$ on RHS of (A) we get
3rd approximate solution as;

$$x^{(3)} = 2.492$$

$$y^{(3)} = 3.685$$

$$z^{(3)} = 1.937$$

Similarly,

4th approximate solution is

$$x^{(4)} = 2.401$$

$$y^{(4)} = 3.545$$

$$z^{(4)} = 1.95$$

5th approximate solution is

$$x^{(5)} = 2.432$$

$$y^{(5)} = 3.583$$

$$z^{(5)} = 1.927$$

6th approximate solution is

$$x^{(6)} = 2.423$$

$$y^{(6)} = 3.57$$

$$z^{(6)} = 1.926$$

7th approximate solution is

$$x^{(7)} = 2.426$$

$$y^{(7)} = 3.574$$

$$z^{(7)} = 1.926$$

Here,
6th + 7th iteration
give same values
for x, y, z upto
2 decimal places.
Thus, solution is
 $x = 2.426$, $y = 3.574$,
 $z = \underline{\underline{1.926}}$

Q Find the iterative solution of the given system using Gauss Jacobi method.

$$6x - 3y + z = 11$$

$$2x + y - 8z = -15$$

$$x - 7y + z = 10$$

sol: The given system of equations is not diagonally dominant. Rearranging the order, we get the corresponding ^{*}diagonally dominated system as,

$$6x - 3y + z = 11 \text{ --- ①} \quad |6| > |-3| + |1| \quad (*)$$

$$x - 7y + z = 10 \text{ --- ②} \quad |1| > |-7| + |1|$$

$$2x + y - 8z = -15 \text{ --- ③} \quad |-8| > |2| + |1|$$

$$\left. \begin{array}{l} \text{Form ① : } x = \frac{1}{6} [11 + 3y - z] \\ \text{Form ② : } y = \frac{1}{-7} [10 - x - z] \\ \text{Form ③ : } z = \frac{1}{-8} [-15 - 2x - y] \end{array} \right\} \text{ (A)}$$

we start with initial approximation $x^{(0)} = y^{(0)} = z^{(0)} = 0$

Substituting $x^{(0)} = y^{(0)} = z^{(0)} = 0$ on the RHS of (A),

$$\left. \begin{array}{l} x^{(1)} = \frac{1}{6} [11 + 3 \times 0 - 0] = \frac{11}{6} \\ y^{(1)} = \frac{1}{-7} [10 - 0 - 0] = -\frac{10}{7} \\ z^{(1)} = \frac{1}{-8} [-15 - 2 \times 0 - 0] = \frac{15}{8} \end{array} \right\} \text{ 1st approximate solution.}$$

Substituting $x^{(1)}, y^{(1)}, z^{(1)}$ on the RHS of (A), we get
2nd approximate solution

$$x^{(2)} = \frac{1}{6} \left[11 + 3 \times \frac{10}{7} - \frac{15}{8} \right] = 0.8064$$

$$y^{(2)} = \frac{1}{-7} \left[10 - \frac{11}{6} - \frac{15}{8} \right] = -1.0454$$

$$z^{(2)} = \frac{1}{-8} \left[-15 - 2 \times \frac{11}{6} - \frac{10}{7} \right] = 1.9459$$

Substituting $x^{(2)}, y^{(2)}, z^{(2)}$ on the RHS of (A), we get
3rd approximate solution,

$$x^{(3)} = \frac{1}{6} \left[11 + 3 \times -1.0454 - 1.9459 \right] = 0.9862$$

$$y^{(3)} = \frac{1}{-7} \left[10 - 0.8064 - 1.9459 \right] = -1.0096$$

$$z^{(3)} = \frac{1}{-8} \left[-15 - 2 \times 0.8064 - -1.0454 \right] = 1.9954$$

We repeat the process and we get the ^{4th} solutions as follows

$$x^{(4)} = 0.9959, \quad y^{(4)} = -1.0011, \quad z^{(4)} = 1.9988$$

$$x^{(5)} = 0.9996, \quad y^{(5)} = -1.0001, \quad z^{(5)} = 1.9999$$

$$x^{(6)} = 0.9999, \quad y^{(6)} = -0.9999, \quad z^{(6)} = 2.0$$

$$x^{(7)} = 0.9999, \quad y^{(7)} = -0.9999, \quad z^{(7)} = 2$$

From 6th & 7th approximations, we see that values of x, y, z are correct upto 4 decimal places. Hence solution is,

$$x = 0.9999 \sim 1, \quad y = -0.9999 \sim -1, \quad z = 2$$

Solution of linear system of equations using

Gauss-Seidel method:

This method is a modification of Jacobi method. As before, the system of linear equations, $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ is written as,

$$x = \frac{1}{a_1} [d_1 - b_1y - c_1z] \quad \text{--- (1)}$$

$$\text{Here ; } |a_1| > |b_1| + |c_1|$$

$$y = \frac{1}{b_2} [d_2 - a_2x - c_2z] \quad \text{--- (2)}$$

$$|b_2| > |a_2| + |c_2|$$

$$z = \frac{1}{c_3} [d_3 - a_3x - b_3y] \quad \text{--- (3)}$$

$$|c_3| > |a_3| + |b_3|$$

We start with initial approximation ; $y^{(0)} = z^{(0)} = 0$ and substituting in (1), we get

$$x^{(1)} = \frac{1}{a_1} [d_1 - b_1y^{(0)} - c_1z^{(0)}] = \frac{1}{a_1} [d_1 - b_1 \cdot 0 - c_1 \cdot 0]$$

$$\Rightarrow x^{(1)} = \frac{d_1}{a_1}$$

Now, substituting $x^{(1)} = \frac{d_1}{a_1}$, $z^{(0)} = 0$ in (2), we get

$$y^{(1)} = \frac{1}{b_2} [d_2 - a_2x^{(1)} - c_2z^{(0)}] = \frac{1}{b_2} [d_2 - a_2 \cdot \frac{d_1}{a_1}]$$

Next, substituting $x^{(1)}$, $y^{(1)}$ in (3), we get

$$z^{(1)} = \frac{1}{c_3} [d_3 - a_3x^{(1)} - b_3y^{(1)}]$$

Thus, $x^{(1)}$, $y^{(1)}$, $z^{(1)}$ are 1st approximate solution.

This process is continued till the values of x, y, z are obtained to the desired degree of accuracy.

Q2 Solve using Gauss Seidel Method.

$$8x - 3y + 2z = 20$$

$$6x + 3y + 12z = 35$$

$$4x + 11y - z = 33$$

Sol:

Before we start the iteration process we check the condition of convergence. Here diagonal elements are not dominant. So rearranging the order, we write

$$8x - 3y + 2z = 20 \quad \text{--- (1)}$$

$$4x + 11y - z = 33 \quad \text{--- (2)}$$

$$6x + 3y + 12z = 35 \quad \text{--- (3)}$$

Here; condition for convergence satisfies.

$$|8| > |-3| + |2|$$

$$|11| > |4| + |-1|$$

$$|12| > |6| + |3|$$

$$\text{From (1): } x = \frac{1}{8} [20 + 3y - 2z] \quad \text{--- (4)}$$

$$\text{From (2): } y = \frac{1}{11} [33 - 4x + z] \quad \text{--- (5)}$$

$$\text{From (3): } z = \frac{1}{12} [35 - 6x - 3y] \quad \text{--- (6)}$$

Starting with initial approximation $y^{(0)} = z^{(0)} = 0$, in RHS of (4) we get

$$x^{(1)} = \frac{1}{8} [20 + 0 - 0] = \frac{20}{8} = 2.5$$

Putting $x^{(1)} = 2.5$, $z^{(0)} = 0$ in RHS of (5), we get

$$y^{(1)} = \frac{1}{11} [33 - 4 \times 2.5 + 0] = 2.090909$$

Putting $x^{(1)} = 2.5$, $y^{(1)} = 2.090909$ in RHS of (6), we get

$$z^{(1)} = \frac{1}{12} [35 - 6 \times 2.5 - 3 \times 2.090909] = 1.1439394$$

2nd approximation

Putting $y^{(1)}$, $z^{(1)}$ in RHS of (4), we get

$$x^{(2)} = \frac{1}{8} [20 + 3y^{(1)} - 2z^{(1)}] = 2.9981$$

Putting $x^{(2)}$, $z^{(1)}$ in RHS of (5), we get

$$y^{(2)} = \frac{1}{11} [33 - 4x^{(2)} + z^{(1)}] = 2.01377$$

Putting $x^{(2)}$, $y^{(2)}$ in RHS of (6), we get

$$z^{(2)} = \frac{1}{12} [35 - 6x^{(2)} - 3y^{(2)}] = 0.91417$$

3rd approximation

$$x_3 = 3.02662$$

$$y_3 = 1.98252$$

$$z_3 = 0.907726$$

4th approximation

$$x_4 = 3.016512$$

$$y_4 = 1.985607$$

$$z_4 = 0.9120088$$

5th approximation

$$x_5 = 3.0166005$$

$$y_5 = 1.9859643$$

$$z_5 = 0.9118753$$

6th approximation

$$x_6 = 3.0167678$$

$$y_6 = 1.9858913$$

$$z_6 = 0.9118099$$

7th approximation

$$x_7 = 3.0167568$$

$$y_7 = 1.9858894$$

$$z_7 = 0.9118159$$

Since at the 6th & 7th approximations, the values of x, y, z are same to four decimal places, we stop the process.

Hence solution is: $x = 3.0167$,
 $y = 1.9858$,
 $z = 0.9118$ //

Q9 Solve the given system of equations using Gauss Seidel method:

$$2x_1 - x_2 = 7, \quad -x_1 + 2x_2 - x_3 = 1, \quad -x_2 + 2x_3 = 1$$

Q1: Given: $2x_1 - x_2 + 0x_3 = 7$ — (1)

$$-x_1 + 2x_2 - x_3 = 1$$
 — (2)

$$0x_1 - x_2 + 2x_3 = 1$$
 — (3)

Here:

$$|2| > |-1| + |0|$$

$$|2| > |-1| + |-1| \quad \boxed{+ \frac{\sqrt{}}{2}}$$

$$|2| > |0| + |-1|$$

Here the diagonal coefficients are dominant. OR condition for convergence satisfies.

Now, from (1): $x_1 = \frac{1}{2} [7 + x_2]$ — (4)

from (2): $x_2 = \frac{1}{2} [1 + x_1 + x_3]$ — (5)

from (3): $x_3 = \frac{1}{2} [1 + x_2]$ — (6)

Starting with $x_2^{(0)} = x_3^{(0)} = 0$, and substituting on RHS of (4), we get

$$x_1^{(1)} = \frac{1}{2} [7 + 0] = \frac{7}{2} = 3.5$$

Putting $x_1^{(1)} = 3.5$, $x_3^{(0)} = 0$ in ^{RHS of} (5), we get

$$x_2^{(1)} = \frac{1}{2} [1 + 3.5 + 0] = 2.25$$

Putting $x_1^{(1)} = 3.5$, $x_2^{(1)} = 2.25$ in RHS of (6), we get

$$x_3^{(1)} = \frac{1}{2} [1 + 2.25] = 1.625$$

2nd approximation:

Put $x_2^{(1)}$ on RHS of (4) we get

$$x_1^{(2)} = \frac{1}{2} [7 + 2 \cdot 25] = 4.625$$

Put $x_1^{(2)} = 4.625$, $x_3^{(1)} = 1.625$ on RHS of (5), we get

$$x_2^{(2)} = \frac{1}{2} [1 + 4 \cdot 625 + 1 \cdot 625] = 3.625$$

Put $x_2^{(2)}$ on RHS of (6), we get

$$x_3^{(2)} = \frac{1}{2} [1 + 3 \cdot 625] = 2.3125$$

3rd approximation:

Put $x_2^{(2)}$ on RHS of (4), we get

$$x_1^{(3)} = \frac{1}{2} [7 + 3 \cdot 625] = 5.3125$$

Put $x_1^{(3)}$, $x_3^{(2)}$ on RHS of (5), we get

$$x_2^{(3)} = \frac{1}{2} [1 + 5 \cdot 3125 + 2 \cdot 3125] = 4.3125$$

Put $x_2^{(3)}$ on RHS of (6), we get

$$x_3^{(3)} = \frac{1}{2} [1 + 4 \cdot 3125] = 2.65625$$

4th approximation:

$$x_1^{(4)} = \frac{1}{2} [7 + x_2^{(3)}] = \frac{1}{2} [7 + 4 \cdot 3125] = 5.65625$$

$$x_2^{(4)} = \frac{1}{2} [1 + x_1^{(4)} + x_3^{(3)}] = \frac{1}{2} [1 + 5 \cdot 65625 + 4 \cdot 3125] = 4.65625$$

$$x_3^{(4)} = \frac{1}{2} [1 + x_2^{(4)}] = \frac{1}{2} [1 + 4 \cdot 65625] = 2.828125$$

Similarly, 5th approximation:

$$x_1^{(5)} = 5.8281, \quad x_2^{(5)} = 4.8281, \quad x_3^{(5)} = 2.9106$$

6th iteration/approximation

$$x_1^{(6)} = 5.9140, \quad x_2^{(6)} = 4.9140, \quad x_3^{(6)} = 2.9570$$

7th approximation

$$x_1^{(7)} = 5.9570, \quad x_2^{(7)} = 4.9570, \quad x_3^{(7)} = 2.9785$$

Since at the 6th & 7th approximations, the values of x_1, x_2, x_3 are same upto one decimal place, we stop the process. Hence the solution is,

$$x_1 = 5.957, \quad x_2 = 4.957, \quad x_3 = 2.978$$

Home Work

KTU

Q1

Using Gauss-Seidel method, solve the following system of equations,

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25.$$

KTU

Q1

Apply Gauss-Seidel method to solve the following system of equations;

$$4x_1 - x_2 - x_3 = 3$$

$$-2x_1 + 6x_2 + x_3 = 9$$

$$-x_1 + x_2 + 7x_3 = -6.$$

Q Use Gauss-Seidel method solve the system :

$$5x_1 - x_2 = 9, \quad -x_1 + 5x_2 - x_3 = 4, \quad -x_2 + 5x_3 = -6$$

sol:

Given:

$$5x_1 - x_2 + 0x_3 = 9 \quad \text{--- (1)}$$

$$-x_1 + 5x_2 - x_3 = 4 \quad \text{--- (2)}$$

$$0x_1 - x_2 + 5x_3 = -6 \quad \text{--- (3)}$$

$$|5| > |-1| + |0|$$

$$|5| > |-1| + |-1|$$

$$|5| > |0| + |-5|$$

The given system of equations are diagonally dominant.

$$\text{From (1), } x_1 = \frac{1}{5} [9 + x_2] \quad \text{--- (4)}$$

$$\text{From (2), } x_2 = \frac{1}{5} [4 + x_1 + x_3] \quad \text{--- (5)}$$

$$\text{From (3), } x_3 = \frac{1}{5} [-6 + x_2] \quad \text{--- (6)}$$

In RHS of (4), putting $x_2^{(0)} = 0$,

$$x_1^{(1)} = \frac{1}{5} [9 + 0] = \frac{9}{5}$$

In RHS of (5), putting $x_1^{(1)} = \frac{9}{5}$, $x_3^{(0)} = 0$,

$$x_2^{(1)} = \frac{1}{5} \left[4 + \frac{9}{5} + 0 \right] = \frac{29}{25}$$

In RHS of (6), putting $x_2^{(1)} = \frac{29}{25}$,

$$x_3^{(1)} = \frac{1}{5} \left[-6 + \frac{29}{25} \right] = -\frac{121}{125}$$

2nd iteration

$$x_1^{(2)} = \frac{1}{5} \left[9 + \frac{29}{25} \right] = 2.032$$

$$x_2^{(2)} = \frac{1}{5} \left[4 + 2.032 + \frac{-121}{125} \right] = 1.0128$$

$$x_3^{(2)} = \frac{1}{5} [-6 + 1.0128] = -0.99744$$

3rd iteration solution

$$x_1^{(3)} = 2.00256$$

$$x_2^{(3)} = 0.868032$$

$$x_3^{(3)} = -1.0263936$$

4th iteration solution

$$x_1^{(4)} = 2.00205$$

$$x_2^{(4)} = 1.00008$$

$$x_3^{(4)} = -0.99998$$

5th iteration solution

$$x_1^{(5)} = 1.99999$$

$$x_2^{(5)} = 0.99999$$

$$x_3^{(5)} = -1.000$$

approximate solution is,

$$x_1 = 2, \quad x_2 = 1, \quad x_3 = -1$$

Curve Fitting: It is the method of finding equation of a curve or straight line that approximates a given set of data.

In many branches of engineering and science, it is required to express the data obtained from observations in the form of some relation connecting the two variables involved. For this purpose, we first plot the corresponding values of the given data say (x_i, y_i) , $i = 1, 2, \dots, n$ on graph paper. A smooth curve can be drawn to pass through near the plotted points. Such a curve is called an approximating curve. The constants occurring in the equation of approximating curve can be found out by the method of least squares.

Method of least squares :

Let $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be a given set of 'n' data points. Let $d_i = y_i - f(x_i)$ denote the deviation which may be +ve, -ve or zero. 'Method of least squares' states that, of all the curves approximating a given set of data points, the curve having the least or minimum sum of squares of the deviations is the "best fitting" curve. If the curve is a straight line fitted according to least squares method, then it is known as least square straight line, if it is a parabola, it is known as least squares parabola etc.

KTU

Fitting of straight line by least square method:

(*) For a given set of points say, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

let the equation of straight line be $y = ax + b$. — (1)

(*) To find the values of constants 'a' and 'b' we use normal equations, given by :-

$$\begin{aligned}\sum y &= a \sum x + nb \\ \sum xy &= a \sum x^2 + b \sum x\end{aligned}$$

On solving these two normal equations, we get values of 'a' and 'b', which is substituted in (1) will give us equation of straight line required.

KTU

Fitting of Parabola by least square method:

(*) For a given set of points say, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

let the equation of parabola be $y = ax^2 + bx + c$ — (1)

* To find the values of constants a, b, c we use normal equations, given by :-

$$\begin{aligned}\sum y &= a \sum x^2 + b \sum x + nc \\ \sum xy &= a \sum x^3 + b \sum x^2 + c \sum x \\ \sum x^2 y &= a \sum x^4 + b \sum x^3 + c \sum x^2\end{aligned}$$

On solving these 3 normal equations, we get values of a, b, c, which when substitute in (1), we get required equation of parabola.

- Q By the method of least squares, find the straight line that best fits the following data:

$$\begin{array}{l} x: 1 \quad 2 \quad 3 \quad 4 \quad 5 \\ y: 14 \quad 27 \quad 40 \quad 55 \quad 68 \end{array}$$

50. Let the equation of straight line of the best fit be

$$y = ax + b \text{ ——— ①}$$

We know, normal equations are,

$$\sum y = a \sum x + nb$$

$$\sum xy = a \sum x^2 + b \sum x$$

The values of $\sum x$, $\sum y$, $\sum xy$, $\sum x^2$ are calculated as shown below:

x	y	x^2	xy
1	14	1	14
2	27	4	54
3	40	9	120
4	55	16	220
5	68	25	340
$\sum x = 15$	$\sum y = 204$	$\sum x^2 = 55$	$\sum xy = 748$

Normal equations become:

$$\begin{array}{l} 204 = a \cdot 15 + b \cdot 5 \\ 748 = a \cdot 55 + b \cdot 15 \end{array} \quad \left. \begin{array}{l} 15a + 5b = 204 \\ 55a + 15b = 748 \end{array} \right\} \text{OR} \quad \left. \begin{array}{l} 15a + 5b = 204 \\ 55a + 15b = 748 \end{array} \right\} \begin{array}{l} \text{On solving,} \\ a = 13.6, b = 0 \end{array}$$

Substituting a & b in ①, $y = \underline{13.6x}$ is the equ. of best fit straight line.

Q Fit a parabola $y = ax^2 + bx + c$ to the data using the method of least squares.

x	10	12	15	23	20
y	14	17	23	25	21

Sol: $y = ax^2 + bx + c$ ——— ① is the equ. of required P.B.

Normal equations are;

$$\begin{aligned} \sum y &= a \sum x^2 + b \sum x + nc \\ \sum xy &= a \sum x^3 + b \sum x^2 + c \sum x \\ \sum x^2 y &= a \sum x^4 + b \sum x^3 + c \sum x^2 \end{aligned}$$

The values of $\sum x$, $\sum y$, $\sum xy$ etc are found out from the table below:

x	y	x^2	x^3	x^4	xy	x^2y
10	14	100	1000	10000	140	1400
12	17	144	1728	20736	204	2448
15	23	225	3375	50625	345	5175
23	25	529	12167	279841	575	13225
20	21	400	8000	160000	420	8400
$\sum x = 80$	$\sum y = 100$	$\sum x^2 = 1398$	$\sum x^3 = 26270$	$\sum x^4 = 521202$	$\sum xy = 1684$	$\sum x^2y = 30648$

Substituting the values obtained from table into the normal equations, we get

$$100 = a(1398) + b(80) + 5c$$

$$1684 = a(26270) + b(1398) + c(80)$$

$$30648 = a(521202) + b(26270) + c(1398)$$

Solving these equations, we get $a = -0.07$, $b = 3.03$ and $c = -8.89$. Substituting these values in ① we get equ. of P.B as,

$$y = -0.07x^2 + 3.03x - 8.89$$

Q. Fit a straight line for the following data by the method of least squares. Also find y at $x=4$ using straight line eqn.

X: 1 2 3 4 5 6
Y: 6 4 3 5 4 2

Sol: Let $y = ax + b$ ——— ①

Normal equations: $\sum y = a \sum x + nb$

$$\sum xy = a \sum x^2 + b \sum x$$

x	y	x^2	$\sum xy$
1	6	1	6
2	4	4	8
3	3	9	9
4	5	16	20
5	4	25	20
6	2	36	12
$\sum x = 21$	$\sum y = 24$	$\sum x^2 = 91$	$\sum xy = 75$

Substituting the values from table to normal eqns,

$$24 = a(21) + b(6)$$

$$75 = a(91) + b(21)$$

Solving, we get $a = -0.514$, $b = 5.8$.

$\therefore$ ① becomes, $y = -0.514x + 5.8$ which is the equations of straight line fitting to the given data.

* y at $x=4$ is, $y = -0.514 \times 4 + 5.8 = \underline{\underline{3.744}}$

Q. Fit a least square quadratic curve to the following data. Also estimate $y(2.4)$

$x: 1 \quad 2 \quad 3 \quad 4$

$y: 1.7 \quad 1.8 \quad 2.3 \quad 3.2$

Sol: Let the quadratic curve be, $y = ax^2 + bx + c$ — (1)

Normal equations are,

$$\sum y = a \sum x^2 + b \sum x + nc$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$$

x	y	x^2	x^3	x^4	xy	x^2y
1	1.7	1	1	1	1.7	1.7
2	1.8	4	8	16	3.6	7.2
3	2.3	9	27	81	6.9	20.7
4	3.2	16	64	256	12.8	51.2
$\sum x = 10$	$\sum y = 9$	$\sum x^2 = 30$	$\sum x^3 = 100$	$\sum x^4 = 354$	$\sum xy = 25$	$\sum x^2y = 80.8$

Substituting the values from the table in normal eqns.

$$\begin{cases} 9 = a(30) + b(10) + 4c \\ 25 = a(100) + b(30) + c(10) \\ 80.8 = a(354) + b(100) + c(30) \end{cases}$$

Solving these equations,
 $a = 0.2, b = 0.5, c = 2$

Thus, (1) becomes, $y = 0.2x^2 - 0.5x + 2$ is the required quadratic curve. Now, $y(2.4) = 0.2(2.4)^2 - 0.5 \times 2.4 + 2 = 1.952$

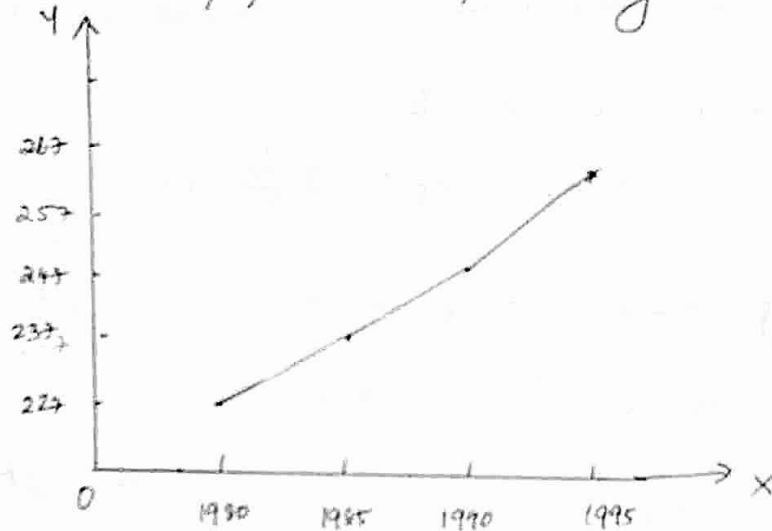
- Q. The table below gives the estimated population of a country (in millions) for during 1980 - 1995.

Year	1980	1985	1990	1995
population	227	237	249	262

Plot a graph of this data and fit an appropriate curve to the data using the method of least squares.

Hence predict the population for the year 2010.

sd:



Graphs plotted for the given data can be approximated to a straight line.

Let $y = ax + b$ — (1) be the eqn. of straight line fitting to the given data.

Normal equations are,

$$\sum y = a \sum x + nb$$

$$\sum xy = a \sum x^2 + b \sum x$$

To find $\sum y$, $\sum x$, $\sum xy$, $\sum x^2$, we construct a table as shown below

x	y	x^2	xy
1980	227	3920400	449460
1985	237	3940225	470445
1990	249	3960100	495510
1995	262	3980025	522690
$\Sigma x = 7950$	$\Sigma y = 975$	$\Sigma x^2 = 15800750$	$\Sigma xy = 1938105$

Substituting the values in normal equations, we get

$$975 = a(7950) + 4b$$

$$1938105 = a(15800750) + b(7950)$$

Solving these two equations, we get

$$a = 2.34, \quad b = -4407$$

Substituting 'a' and 'b' in (1), we get

$$\underline{\underline{y = 2.34x - 4407}} \text{ is the required eqn. of straight line.}$$

Population for the year 2010 is

$$\begin{aligned} y &= 2.34 \times 2010 - 4407 \\ &= \underline{\underline{296.4 \text{ millions}}} \end{aligned}$$

- Q Using the method of least square, fit a straight line to the points $(-1.0, 1.0)$, $(-0.1, 1.099)$, $(0.2, 0.808)$, $(1, 1)$

Sol: Let the equ. of straight line of the best fit be,

$$y = ax + b \text{ --- ①}$$

Normal equations are,

$$\sum y = a \sum x + n b$$

$$\sum xy = a \sum x^2 + b \sum x$$

The values of $\sum y$, $\sum x$, $\sum xy$, $\sum x^2$ are calculated as shown below:

x	y	x^2	xy
-1	1	1	-1
-0.1	1.099	0.01	-0.1099
0.2	0.808	0.04	0.1616
1	1	1	1
$\sum x = 0.1$		$\sum y = 3.907$	$\sum x^2 = 2.05$
			$\sum xy = 0.0517$

Normal equations becomes,

$$3.907 = a(0.1) + 4b$$

$$0.0517 = a(2.05) + b(0.1)$$

On solving, $a = 0.9773$ &
 $b = -0.0224$

$$\therefore \text{①} \Rightarrow y = \underline{0.9773x - 0.0224}$$

- Q1) Employ the method of least squares to fit a parabola for the data: $(-1, 2), (0, 0), (0, 1), (1, 2)$

Sol: Let the eqn. of P.B be, $y = ax^2 + bx + c$ — (1)

Normal equations are,

$$\sum y = a \sum x^2 + b \sum x + nc$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x$$

$$\sum x^2 y = a \sum x^4 + b \sum x^3 + c \sum x^2$$

The values of $\sum x, \sum y, \sum xy, \dots$ are found out from the following table as shown below:

x	y	x^2	x^3	x^4	xy	x^2y
-1	2	1	-1	1	-2	2
0	0	0	0	0	0	0
0	1	0	0	0	0	0
1	2	1	1	1	2	2
$\sum x = 0$	$\sum y = 5$	$\sum x^2 = 2$	$\sum x^3 = 0$	$\sum x^4 = 2$	$\sum xy = 0$	$\sum x^2y = 4$

Substituting the values obtained from table into the normal equations, we get

$$5 = a(2) + b(0) + 4c \Rightarrow 5 = 2a + 4c$$

$$0 = a(0) + b(2) + c(0) \Rightarrow 0 = 2b \Rightarrow b = 0$$

$$4 = a(2) + b(0) + c(2) \Rightarrow 4 = 2a + 2c$$

Solving these equations, we get $a = \frac{1}{2}, c = \frac{3}{2}, b = 0$

$$\therefore \text{Eqn. (1) becomes, } y = \frac{1}{2}x^2 + 0x + \frac{3}{2}$$

$$\Rightarrow \underline{\underline{y = \frac{x^2}{2} + \frac{3}{2}}} \text{ is the eqn. of P.B of best fit.}$$

Home work

Q

KTU

Using the method of least squares fit a straight line of the form $y = ax + b$ to the following set of ordered pairs (x, y) : $(2, 4)$, $(3, 5)$, $(5, 7)$, $(7, 10)$, $(9, 15)$

Q

KTU

Explain the principle of least squares for determining a line of best fit to a given data.

①

Solution of ordinary differential equations:

In analytical methods, we get the exact solution of the differential equation in the form $y=f(x)$, where in numerical methods we get pointwise solutions $y=y_1, y_2, y_3, \dots$ corresponding to $x=x_1, x_2, \dots$ using suitable iterative/recursive formula. The difference between the computed value y_i and the true value $f(x_i)$ is termed as truncation error at $x=x_i$.

Euler's method, Runge-Kutta method of 2nd order, Runge-Kutta method of 4th order (classical method) & Adams Moulton predictor-corrector method are some of the numerical methods to solve i.v.p.

Euler's Method

To solve the o.d.e,

$$y' = \frac{dy}{dx} = f(x, y) \quad , \quad y(x_0) = y_0$$

we use the recursive formula,

$$y_{n+1} = y_n + h f(x_n, y_n)$$

h=step-size

to get point-wise solutions.

2] Solve : $\frac{dy}{dx} = \frac{x-y}{2}$, $y(0)=1$ using Euler's method over the interval $[0, 3]$ with step size $h=0.5$. Also find $y(2.5)$. (3)

Sol:

Given: $\frac{dy}{dx} = \frac{x-y}{2} \Rightarrow f(x, y) = \frac{x-y}{2}$

$y(0)=1 \equiv y(x_0)=y_0$

$\therefore x_0=0, y_0=1$

Euler's method

$$\begin{cases} y' = \frac{dy}{dx} = f(x, y), y(x_0) = y_0 \\ \Rightarrow y_{n+1} = y_n + h f(x_n, y_n) \end{cases}$$

Since $h=0.5$, x -values are,

$x_0=0, x_1=0.5, x_2=1, x_3=1.5, x_4=2, x_5=2.5, x_6=3$

Corresponding to each of these x -values, we obtain $y_0, y_1, \dots$ which are the pointwise solution to the given o.d.e.

Now, $y_{n+1} = y_n + h f(x_n, y_n)$

$\Rightarrow y_{n+1} = y_n + 0.5 \left(\frac{x_n - y_n}{2} \right)$

$= y_n + \frac{1}{2} \left(\frac{x_n - y_n}{2} \right)$

$= y_n + \frac{x_n}{4} - \frac{y_n}{4}$

$= \frac{3}{4} y_n + \frac{1}{4} x_n$

$f(x, y) = \frac{x-y}{2}$

$f(x_n, y_n) = \frac{x_n - y_n}{2}$

Thus, we have

$$y_{n+1} = \frac{3}{4} y_n + \frac{1}{4} x_n \quad \text{--- (1)}$$

Putting $n=0, 1, 2, 3, \dots$ in (1) we get

$$n=0 \rightarrow y_1 = \frac{3}{4} y_0 + \frac{1}{4} x_0 = \frac{3}{4} \times 1 + \frac{1}{4} \times 0 = \frac{3}{4} = 0.75$$

$$n=1 \rightarrow y_2 = \frac{3}{4} y_1 + \frac{1}{4} x_1 = \frac{3}{4} \times 0.75 + \frac{1}{4} \times 0.5 = 0.6875$$

$$n=2 \rightarrow y_3 = \frac{3}{4} y_2 + \frac{1}{4} x_2 = \frac{3}{4} \times 0.6875 + \frac{1}{4} \times 1 = 0.7656$$

$$n=3 \rightarrow y_4 = \frac{3}{4} y_3 + \frac{1}{4} x_3 = \frac{3}{4} \times 0.7656 + \frac{1}{4} \times 1.5 = 0.9492$$

$$n=4 \rightarrow y_5 = \frac{3}{4} y_4 + \frac{1}{4} x_4 = \frac{3}{4} \times 0.9492 + \frac{1}{4} \times 2 = 1.2119$$

$$n=5 \rightarrow y_6 = \frac{3}{4} y_5 + \frac{1}{4} x_5 = \frac{3}{4} \times 1.2119 + \frac{1}{4} \times 2.5 = 1.5339$$

Given:
 $y(0) = 1$
 $\downarrow$
 $x_0 = 0$ $y_0 = 1$

$h=0.5$

$x_0 = 0$
$x_1 = 0.5$
$x_2 = 1$
$x_3 = 1.5$
$x_4 = 2$
$x_5 = 2.5$
$x_6 = 3$

Thus, we have pointwise solution as follows:

$$y(0) = y_0 = 1 \text{ (given)}$$

$$y(0.5) = y_1 = 0.75$$

$$y(1) = y_2 = 0.6875$$

$$y(1.5) = y_3 = 0.7656$$

$$y(2) = y_4 = 0.9492$$

$$y(2.5) = y_5 = 1.2119$$

$$y(3) = y_6 = 1.5339$$

This, $\underline{\underline{y(2.5) = 1.2119}}$.

* **NOTE:** In the question it is given that the interval is $[0, 3]$. So we stop the process when x we get $y(3)$. i.e. value of y when $x=3$.

Q] Given the initial value problem $y' = y + x$, $y(0) = 0$.
Find $y(0.2)$ and $y(0.4)$ using Euler's method. ($h = 0.2$)

Sol:

Given: $y' = y + x \Rightarrow f(x, y) = y + x$

$y(0) = 0 \Rightarrow y(x_0) = y_0$

$\therefore x_0 = 0, y_0 = 0$

Euler's method

$$\left\{ \begin{aligned} y' &= \frac{dy}{dx} = f(x, y), y(x_0) = y_0 \\ \Rightarrow y_{n+1} &= y_n + h f(x_n, y_n) \end{aligned} \right.$$

Since $h = 0.2$, x -values are

$x_0 = 0, x_1 = 0.2, x_2 = 0.4, \dots$

Now, $y_{n+1} = y_n + h f(x_n, y_n)$

$= y_n + 0.2 (y_n + x_n)$

$= y_n + 0.2 y_n + 0.2 x_n$

$= (1.2) y_n + (0.2) x_n$

$f(x, y) = y + x$

$f(x_n, y_n) = y_n + x_n$

Thus, $y_{n+1} = (1.2) y_n + (0.2) x_n$ ——— ①

Putting $n = 0, 1, 2, \dots$

Put $n = 0$ in ① $\rightarrow y_1 = (1.2) y_0 + (0.2) x_0 = 0$

$n = 1 \rightarrow y_2 = (1.2) y_1 + (0.2) x_1 = 1.2 \times 0 + (0.2) \times (0.2)$
 $= 0.04$

$n = 2 \rightarrow y_3 = (1.2) y_2 + (0.2) x_2 = (1.2) (0.04) + (0.2) (0.4)$
 $= 0.128$

Now: To find $y(0.2)$ & $y(0.4)$

$y(0.2) = y \text{ at } x_1 = y_1 = 0 \Rightarrow y(0.2) = 0$

$y(0.4) = y \text{ at } x_2 = y_2 = 0.04 \Rightarrow y(0.4) = 0.04$

Given:
 $y(0) = 0$
 $\downarrow$
 $x_0 = 0 \quad y_0 = 0$
 Since $h = 0.2$,
 $x_0 = 0$
 $x_1 = 0.2$
 $x_2 = 0.4$

Q Using Euler's method, find an approximate value of y corresponding to $x=1.3$ for the i.v.p

$$\frac{dy}{dx} = x + 2y, \quad y(1) = 1$$

Sol: Given: $\frac{dy}{dx} = x + 2y \Rightarrow f(x, y) = x + 2y$

$$y(1) = 1 \equiv y(x_0) = y_0$$

$$\therefore x_0 = 1, \quad y_0 = 1$$

As we need to find y at $x=1.3$, we take $h=0.1$

Thus, x -values are,

$$x_0 = 1, \quad x_1 = 1.1, \quad x_2 = 1.2, \quad x_3 = 1.3$$

we need to find y at 1.3 i.e. y_3

$$\begin{aligned} \text{Now, } y_{n+1} &= y_n + h f(x_n, y_n) \\ &= y_n + 0.1 (x_n + 2y_n) \\ &= y_n + 0.1 x_n + 0.2 y_n \\ &= 1.2 y_n + 0.1 x_n \end{aligned}$$

$$\begin{aligned} f(x, y) &= x + 2y \\ f(x_n, y_n) &= x_n + 2y_n \end{aligned}$$

$$\text{i.e. } y_{n+1} = 1.2 y_n + 0.1 x_n \quad \text{--- (1)}$$

$$n=0 \rightarrow y_1 = (1.2) y_0 + (0.1) x_0 = 1.2 \times 1 + (0.1) 1 = 1.3$$

$$n=1 \rightarrow y_2 = (1.2) y_1 + (0.1) x_1 = (1.2)(1.3) + (0.1)(1.1) = 1.67$$

$$n=2 \rightarrow y_3 = (1.2) y_2 + (0.1) x_2 = (1.2)(1.67) + (0.1)(1.2) = 2.12$$

Given $y(1) = 1$
 $\swarrow \quad \searrow$
 $x_0 = 1 \quad y_0 = 1$
 $h = 0.1$
 $\left\{ \begin{array}{l} x_0 = 1 \\ x_1 = 1.1 \\ x_2 = 1.2 \\ x_3 = 1.3 \end{array} \right.$

Thus, y at $x=1.3$ i.e. y at $x_3 = \underline{\underline{y_3 = 2.12}}$

NOTE: (*) If take $h=0.2$, x values will be 1, 1.2, 1.4...
 then we won't be able to find y at 1.3

Q Use Euler method to solve $y' = x + xy + y$, $y(0) = 1$ for value of y at $x = 0.15$. (Take $h = 0.05$)

Sol:

Given: $y' = x + xy + y$

$$\Rightarrow f(x, y) = x + xy + y$$

Also, $y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$

Since $h = 0.05$, x -values are,

$$x_0 = 0, x_1 = 0.05, x_2 = 0.10, x_3 = 0.15.$$

we need to find value of y at $x = 0.15$

i.e. we need to find y_3 . For that initially we need to find y_1 & y_2 .
we've: $y_{n+1} = y_n + h f(x_n, y_n)$

$$n=0 \rightarrow y_1 = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.05 f(0, 1)$$

$$= 1 + 0.05 [0 + 0 \times 1 + 1] = \underline{1.05}$$

$$\boxed{f(x, y) = x + xy + y}$$

$$n=1 \rightarrow y_2 = y_1 + h f(x_1, y_1)$$

$$= 1.05 + 0.05 f(0.05, 1.05)$$

$$= 1.05 + 0.05 [0.05 + 0.05 \times (1.05) + 1.05]$$

$$= 1.107625$$

$$n=2 \rightarrow y_3 = y_2 + h f(x_2, y_2)$$

$$= 1.107625 + 0.05 [f(0.10, 1.107625)]$$

$$= 1.107625 + 0.05 [0.10 + 0.1 \times 1.107625 + 1.107625]$$

$$= \underline{1.173544}$$

Thus value of y at $x = 0.15$ is 1.173544

Q Solve the i.v.p $y' + 0.2y = 0$, $y(0) = 5$, $h = 0.2$ using Euler's method taking 3 steps.

Sol: Given: $y' + 0.2y = 0$
 $\Rightarrow y' = -0.2y$
 $\Rightarrow f(x, y) = -0.2y$

Euler's method

$$\left\{ \begin{array}{l} * y' = \frac{dy}{dx} = f(x, y), y(x_0) = y_0 \\ * y_{n+1} = y_n + h f(x_n, y_n) \end{array} \right.$$

Also, $y(0) = 5 \Rightarrow x_0 = 0, y_0 = 5$

As we need to find the solution in 3 steps, we've to find y_1, y_2 and y_3 corresponding to x_1, x_2, x_3 .

Since $h = 0.2$, x -values are,

$$x_0 = 0, x_1 = 0.2, x_2 = 0.4, x_3 = 0.6$$

Now, $y_{n+1} = y_n + h f(x_n, y_n) \quad \text{--- (1)}$

$$n=0 \rightarrow y_1 = y_0 + h f(x_0, y_0) = 5 + 0.2 f(0, 5) = \boxed{f(x, y) = -0.2y}$$
$$= 5 + 0.2(-0.2 \times 5) = \underline{4.8}$$

$$n=1 \rightarrow y_2 = y_1 + h f(x_1, y_1) = 4.8 + 0.2 f(0.2, 4.8)$$
$$= 4.8 + 0.2(-0.2 \times 4.8) = \underline{4.608}$$

$$n=2 \rightarrow y_3 = y_2 + h f(x_2, y_2) = 4.608 + 0.2 f(0.4, 4.608)$$
$$= 4.608 + 0.2[-0.2 \times 4.608]$$
$$= \underline{\underline{4.42368}}$$

Home work

(8)

KTU

[5]

Given the initial value problem, $y' = y + x$, $y(0) = 0$. Find $y(0.1)$ and $y(0.2)$ using Euler method.

[4]

Using Euler's method, obtain the solution of $\frac{dy}{dx} = -y$, $y(0) = 1$. Also find the exact solution and compare the value of y at $x = 0.04$ in both case.

Hint: Since $x_0 = 0$ and we need to find y at $x = 0.04$, we may take $h = 0.01$ or $h = 0.02$.

To get exact solution

$$\begin{aligned}\frac{dy}{dx} = -y &\Rightarrow \frac{dy}{y} = -dx \Rightarrow \log y = -x + \log c \\ &\Rightarrow \log\left(\frac{y}{c}\right) = -x \\ &\Rightarrow \frac{y}{c} = e^{-x} \text{ or } \underline{\underline{y = ce^{-x}}}\end{aligned}$$

Given, $y(0) = 1$.

$$\therefore 1 = c \cdot e^{-0} \Rightarrow c = 1 \quad \therefore \text{Solution is } \underline{\underline{y = e^{-x}}}$$

$$y(\text{at } 0.04) = e^{-0.04} = \underline{\underline{0.9608}}$$

⑨

Runge-Kutta method of 2nd order (Improved Euler method or Modified Euler method)

To solve the o.d.e,

$$y' = \frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

we use the recursive formula,

$$y_{n+1} = y_n + \frac{1}{2} (k_1 + k_2) \quad \text{where}$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f(x_n + h, y_n + k_1)$$

to get pointwise solutions.

Q Apply R-K method of order 2 to approximate value of y when $x=1.1$. Given i.v.p is $\frac{dy}{dx} = 3x+y^2$, $y(1)=1.2$

Sol:

Given:

$$\frac{dy}{dx} = 3x+y^2 \Rightarrow f(x,y) = 3x+y^2$$

$$y(1) = 1.2 \Rightarrow x_0 = 1, y_0 = 1.2$$

we are asked to find value of y when $x=1.1$.

Since, $x_0=1$, taking $h=0.1$

we get $x_1=1.1$. Therefore, we need to find y when $x_1=1.1$ means, we've to find y_1 .

To find y_1 (ie $y(1.1)$)

Putting $n=0$ in equ. ① & ②, we get

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2) \text{ --- ③ where}$$

$$k_1 = h f(x_0, y_0)$$

$$= (0.1) f(1, 1.2)$$

$$= (0.1) [3 \times 1 + (1.2)^2] \because f(x,y) = 3x+y^2$$

$$= (0.1) [3 + 1.44]$$

$$= 0.444$$

$$k_2 = h f(x_0+h, y_0+k_1) = (0.1) f(1+0.1, 1.2+0.444) = (0.1) f(1.1, 1.644)$$

$$= (0.1) [3 \times 1.1 + (1.644)^2] = 0.60027$$

$$\text{Substituting values of } y_0, k_1, k_2 \text{ in ③, } y_1 = 1.2 + \frac{1}{2}(0.444 + 0.60027) = 1.72213 //$$

Q. Given i.v.p, $y' = y - x$, $y(0) = 2$. Find $y(0.2)$ by Runge Kutta method of order 2. (Take $h = 0.1$)

SA. Given: $y' = y - x \Rightarrow f(x, y) = y - x$

$$y(0) = 2 \Rightarrow x_0 = 0, y_0 = 2$$

Since $h = 0.1$, other x values are

$x_1 = 0.1, x_2 = 0.2, \dots$. We need to find

$y(0.2)$ i.e. y_2 . To find y_2 , initially

we need to find y_1 .

To find y_1

Put $n=0$ in ① & ②, we get

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2) \text{ --- ③}$$

$$k_1 = h f(x_0, y_0) = (0.1) f(0, 2) = (0.1)[2 - 0] = \underline{0.2}$$

$$\begin{aligned} k_2 &= h f(x_0 + h, y_0 + k_1) = (0.1) f(0 + 0.1, 2 + 0.2) = (0.1) f(0.1, 2.2) \\ &= (0.1)[2.2 - 0.1] = (0.1)(2.1) = \underline{0.21} \end{aligned}$$

Substituting y_0, k_1, k_2 in ③

$$y_1 = 2 + \frac{1}{2}(0.2 + 0.21) = 2 + \frac{0.41}{2} = \underline{2.205}$$

To find y_2

Put $n=1$ in ① & ②, we get

$$y_2 = y_1 + \frac{1}{2}(k_1 + k_2) \text{ --- ④}$$

$$k_1 = h f(x_1, y_1) = (0.1) f(0.1, 2.205) = (0.1)[2.205 - 0.1] = 0.2105$$

$$\begin{aligned} k_2 &= h f(x_1 + h, y_1 + k_1) = (0.1) f(0.1 + 0.1, 2.205 + 0.2105) \\ &= (0.1) f(0.2, 2.4155) = (0.1)[2.4155 - 0.2] \\ &= 0.22155 \end{aligned}$$

Substituting y_1, k_1, k_2 in ④,

$$y_2 = 2.205 + \frac{1}{2}(0.2105 + 0.22155) = \underline{2.42103}$$

$$\text{i.e. } y(0.2) = \underline{2.42103}$$

R.K of order 2

$$y' = \frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2) \text{ --- ①}$$

$$\left. \begin{aligned} k_1 &= h f(x_n, y_n) \\ k_2 &= h f(x_n + h, y_n + k_1) \end{aligned} \right\} \text{ --- ②}$$

$$\boxed{f(x, y) = y - x}$$

$$\left\{ \begin{aligned} x_0 = 0 &\rightarrow y_0 = 2 \\ x_1 = 0.1 &\rightarrow y_1 = 2.205 \\ x_2 = 0.2 &\rightarrow y_2 = ? \end{aligned} \right.$$

$$\boxed{f(x, y) = y - x}$$

Q. Use Runge-Kutta Method of ⁽¹²⁾ 2nd order to find an approximate value of y when $x=0.02$, where i.v.p is given by $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$

Sol: Given: $\frac{dy}{dx} = x^2 + y \rightarrow f(x, y) = x^2 + y$ R.K method of 2nd order

$$y(0) = 1 \Rightarrow \begin{matrix} x_0 = 0 \\ y_0 = 1 \end{matrix}$$

we need to find y at $x=0.02$:

So let us take $h=0.01$; so that

values of x are,

$$x_0 = 0, x_1 = 0.01, x_2 = 0.02$$

Thus, value of y when $x=0.02$ is y_2 .

To find y_2 , initially we need to find y_1 .

To find y_1 :

Put $n=0$ in (1) & (2); we get

$$y_1 = y_0 + \frac{1}{2}(K_1 + K_2) \quad \text{--- (3)}$$

$$\begin{aligned} K_1 &= h f(x_0, y_0) = 0.01 f(0, 1) \\ &= (0.01) [0^2 + 1] \\ &= \underline{\underline{0.01}} \end{aligned}$$

$$f(x, y) = x^2 + y$$

$$\begin{aligned} K_2 &= h f(x_0 + h, y_0 + K_1) = (0.01) f[0 + 0.01, 1 + 0.01] \\ &= (0.01) f(0.01, 1.01) = (0.01) [(0.01)^2 + 1.01] \\ &= \underline{\underline{0.010}} \end{aligned}$$

Substituting y_0, K_1, K_2 in eqn (3) we get

$$y_1 = 1 + \frac{1}{2}(0.01 + 0.010) = \underline{\underline{1.010}}$$

To find y_2 (ie value of y at $x_2 = 0.02$)

Put $n=1$ in eqn. (1) & (2), we get

$$y_2 = y_1 + \frac{1}{2}(K_1 + K_2) \quad \text{--- (4)}$$

$$K_1 = h f(x_1, y_1)$$

$$= (0.01) f(0.01, 1.010)$$

$$= (0.01) [(0.01)^2 + 1.010]$$

$$= \underline{\underline{0.0101}}$$

$$K_2 = h f(x_1 + h, y_1 + K_1)$$

$$= (0.01) f(0.01 + 0.01, 1.010 + 0.0101)$$

$$= (0.01) f(0.02, 1.0201)$$

$$= (0.01) [(0.02)^2 + 1.0201] \quad \because f(x, y) = x^2 + y$$

$$= \underline{\underline{0.0102}}$$

Substituting y_1, K_1, K_2 in eqn. (4),

$$y_2 = 1.010 + \frac{1}{2}(0.0101 + 0.0102) = 1.0202$$

Thus, $y_2 = 1.0202$ is corresponding to x_2 .

OR

$$\underline{\underline{y(0.02) = 1.0202}}$$

$$x_0 = 0$$

$$x_1 = 0.01$$

$$x_2 = 0.02$$

Given

$$y(0) = 1$$

$$\downarrow$$

$$x_0 = 0$$

$$\searrow$$

$$y_0 = 1$$

Since $h = 0.01$

$$x_1 = 0.01, y_1 = 1.01$$

(we found)

$$f(x, y) = x^2 + y$$

Q Use Runge-Kutta 2nd order method to find the value of y at $x = 0.25$ and $x = 0.5$. Given: $\frac{dy}{dx} = 2xy$, $y(0) = 1$

SA: Given: $\frac{dy}{dx} = 2xy$

$$\Rightarrow f(x, y) = 2xy$$

$$y(0) = 1 \Rightarrow x_0 = 0, y_0 = 1$$

Since we need to find value of y at $x = 0.25$ & 0.50 ,

we take h as, $h = 0.25$. Thus x -values are,

$x_0 = 0, x_1 = 0.25, x_2 = 0.50$. We need to find y at x_1 and y at x_2 . i.e. we have to find y_1 & y_2

To find y_1 (i.e. $y(0.25)$)

Put $n=0$ in ① & ②, then we get

$$y_1 = y_0 + \frac{1}{2}(k_1 + k_2) \quad \text{--- ③ where}$$

$$f(x, y) = 2xy$$

$$k_1 = h f(x_0, y_0) = 0.25 f(0, 1) = 0.25 [2 \times 0 \times 1] = 0$$

$$k_2 = h f(x_0 + h, y_0 + k_1) = 0.25 f(0.25, 1 + 0)$$

$$= (0.25) [2 \times 0.25 \times 1]$$

$$= \underline{\underline{0.125}}$$

$\therefore$ ③ becomes,

$$y_1 = 1 + \frac{1}{2}(0 + 0.125) = \underline{\underline{1.0625}} \quad \text{i.e. } y(0.25) = \underline{\underline{1.0625}}$$

(15)

To find y_2 (ie $y(0.50)$)

Put $n=1$ in ① & ② we get

$$y_2 = y_1 + \frac{1}{2}(k_1 + k_2) \quad \text{--- ④}$$

$$k_1 = h f(x_1, y_1)$$

$$= 0.25 f(0.25, 1.0625)$$

$$= 0.25 [2 \times 0.25 \times 1.0625]$$

$$= \underline{\underline{0.1328}}$$

$f(x, y) = 2xy$

$$k_2 = h f(x_1 + h, y_1 + k_1)$$

$$= 0.25 f(0.25 + 0.25, 1.0625 + 0.1328)$$

$$= 0.25 f(0.5, 1.1953)$$

$$= 0.25 [2 \times 0.5 \times 1.1953]$$

$$= \underline{\underline{0.2988}}$$

∴ ④ becomes, $y_2 = 1.0625 + \frac{1}{2}(0.1328 + 0.2988)$

$$= \underline{\underline{1.2783}}$$

ie value of y at $x = 0.5$ (or)

$$\underline{\underline{y(0.5) = 1.2783}}$$

Runga-Kutta method of ⁽¹⁶⁾ 4th order

To solve the o.d.e,

$$y' = \frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

we use the recursive formula,

$$y_{n+1} = y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad \text{where}$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_n + h, y_n + k_3)$$

to get pointwise solutions.

- Q Apply R.K method of 4th order (classical R.K method) to find an approximate value for $y(0.2)$ from the i.v.p $y' = x + y, y(0) = 1$ (Take $h = 0.2$).

Sol:

Given: $y' = x + y \rightarrow f(x, y) = x + y$
 $y(0) = 1 \Rightarrow x_0 = 0$ and $y_0 = 1$

Since $h = 0.2$, values of x are, $x_0 = 0, x_1 = 0.2$.

we need to find $y(0.2)$
 i.e. we need to find y_1 .

To find y_1 i.e. $y(0.2)$

R.K method of 4th order

$$y' = \frac{dy}{dx} = f(x, y), y(x_0) = y_0 \quad \text{--- (1)}$$

$$y_{n+1} = y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad \text{--- (2)}$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_n + h, y_n + k_3)$$

Put $n=0$ in (2) & (3), we get

$$y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad \text{--- (4)}$$

$$k_1 = h f(x_0, y_0) = 0.2 f(0, 1) = 0.2 (0 + 1) = \underline{0.2}$$

$$\boxed{f(x, y) = x + y}$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= 0.2 f\left(0 + \frac{0.2}{2}, 1 + \frac{0.2}{2}\right)$$

$$= 0.2 f(0.1, 1.1)$$

$$= 0.2 (0.1 + 1.1) = 0.2 (1.2) = \cancel{0.4} \underline{0.24}$$

(18)

$$\begin{aligned}
 K_3 &= h f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right) \\
 &= 0.2 f\left(0 + \frac{0.2}{2}, 1 + \frac{0.24}{2}\right) \\
 &= 0.2 f(0.1, 1.12) \\
 &= 0.2 [0.1 + 1.12] \\
 &= 0.2 \times 1.22 \\
 &= \underline{\underline{0.244}}
 \end{aligned}$$

$$\boxed{f(x, y) = x + y}$$

$$\begin{aligned}
 K_4 &= h f(x_0 + h, y_0 + K_3) \\
 &= 0.2 f(0 + 0.2, 1 + 0.244) \\
 &= 0.2 f(0.2, 1.244) \\
 &= 0.2 [0.2 + 1.244] \\
 &= 0.2888
 \end{aligned}$$

Substituting the values of y_0, K_1, K_2, K_3 & K_4 in (4) we get,

$$\begin{aligned}
 y_1 &= 1 + \frac{1}{6} [0.2 + 2 \times 0.24 + 2 \times 0.244 + 0.2888] \\
 &= 1.2428
 \end{aligned}$$

Thus, when $x = 0.2$, $y = 1.2428$

$$\text{or } \underline{\underline{y(0.2) = 1.2428 \approx 1.24}}$$

Q

Solve: $\frac{dy}{dx} = xy + y^2$, $y(0)=1$ for finding solution at $x=0.1$ and $x=0.2$ using Runge-Kutta method of order 4. (Choose $h=0.1$)

Sol:

Given: $\frac{dy}{dx} = xy + y^2 \Rightarrow f(x,y) = xy + y^2$

$y(0)=1 \Rightarrow x_0=0$, and $y_0=1$

Since $h=0.1$, x -values are

$x_0=0, x_1=0.1, x_2=0.2$.

We need to find $y(0.1)$ and $y(0.2)$ i.e. y_1 and y_2 .

To find y_1 [i.e. $y(0.1)$]

Put $n=0$ in (2) + (3)

$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ — (4) where

$k_1 = h f(x_0, y_0)$

$= (0.1) f(0, 1)$

$= (0.1) [0 \cdot 1 + 1^2]$

$= 0.1$

$f(x,y) = xy + y^2$

(Given)
 $y(0)=1$
 $x_0=0, y_0=1$

Runge-Kutta method of 4th order

$y' = \frac{dy}{dx} = f(x,y), y(x_0) = y_0$ — (1)

$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ — (2)

$k_1 = h f(x_n, y_n)$

$k_2 = h f(x_n + \frac{h}{2}, y_n + \frac{k_1}{2})$

$k_3 = h f(x_n + \frac{h}{2}, y_n + \frac{k_2}{2})$ — (3)

$k_4 = h f(x_n + h, y_n + k_3)$

(20)

$$K_2 = h f \left[x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2} \right]$$

$$= (0.1) f \left[0 + \frac{0.1}{2}, 1 + \frac{0.1}{2} \right]$$

$$= (0.1) f [0.05, 1.05]$$

$$= (0.1) \left[(0.05 \times 1.05) + (1.05)^2 \right]$$

$$= \underline{\underline{0.1155}}$$

$$f(x, y) = xy + y^2$$

$$K_3 = h f \left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2} \right)$$

$$= (0.1) f \left[0 + \frac{0.1}{2}, 1 + \frac{0.1155}{2} \right]$$

$$= (0.1) f [0.05, 1.05775]$$

$$= (0.1) \left[(0.05)(1.05775) + (1.05775)^2 \right]$$

$$= 0.1171691$$

$$K_4 = h f (x_0 + h, y_0 + K_3)$$

$$= (0.1) f (0 + 0.1, 1 + 0.1171691)$$

$$= (0.1) f [0.1, 1.1171691]$$

$$= (0.1) \left[(0.1)(1.1171691) + (1.1171691)^2 \right]$$

$$= 0.1359784$$

(21)

Substituting the values of y_0, k_1, k_2, k_3, k_4 in in (4)

$$y_1 = 1 + \frac{1}{6}(0.1 + 2 \times 0.1155 + 2 \times 0.1171691 + 0.1359784) \\ = 1.1168861$$

Thus at $x=0.1$, value of $y = y_1 = \underline{\underline{1.1168861}} \approx y$

To find y_2 ie $y(0.2)$

Put $n=1$ in equ. (2) & (3)

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad \text{--- (5) where}$$

$$k_1 = h f(x_1, y_1) = (0.1) f(0.1, 1.1168861) \quad \boxed{f(x,y) = xy + y^2} \\ = (0.1) \left[(0.1)(1.1168861) + (1.1168861)^2 \right] \\ = \underline{\underline{0.1359123}}$$

$$k_2 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = h f\left(0.1 + \frac{0.1}{2}, 1.1168861 + \frac{0.1359123}{2}\right) \\ = h f(0.15, 1.1848423) = 0.1 \left[0.15 \times 1.1848423 + (1.1848423)^2 \right] \\ = \underline{\underline{0.158158}}$$

$$k_3 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.1 \left[f\left(0.1 + \frac{0.1}{2}, 1.1168861 + \frac{0.158158}{2}\right) \right] \\ = (0.1) \left[f(0.15, 1.195965) \right] = (0.1) \left[0.15 \times 1.195965 + (1.195965)^2 \right] \\ = \underline{\underline{0.160973}}$$

$$\begin{aligned}
 K_4 &= h f(x_1+h, y_1+k_3) \stackrel{(22)}{=} (0.1) f(0.1+0.1, 1.1168861+0.160973) \\
 &= (0.1) [f(0.2, 1.277859)] \quad \boxed{f(x,y) = xy + y^2} \\
 &= 0.1888496
 \end{aligned}$$

Substituting the values of y_1, K_1, K_2, K_3 & K_4 in (5),

$$\begin{aligned}
 y_2 &= 1.1168861 + \frac{1}{6} (0.1359123 + 2 \times 0.158158 + \\
 &\quad 2 \times 0.160973 + 0.1888496) \\
 &= 1.2773914
 \end{aligned}$$

ie at $x=0.2$, $y = 1.2773914 \approx y$

Home work:

- Q] Using R.K method of 4th order, obtain the solution at $y(1.3)$ from the i.v.p, $\frac{dy}{dx} = xy^2 - \frac{y}{x}$, $y(1)=1$ correct to 4 decimal places.

KTO

- Q] Use R.K method of fourth order to find $y(0.2)$. Given the i.v.p $\frac{dy}{dx} = \frac{xy}{1+x^2}$, $y(0)=1$ (Take $h=0.1$)

KTO

- Q] Use R.K method of 4th order to compute $y(0.25)$ and $y(0.5)$, Given the i.v.p, $y' = x + xy + y$, $y(0)=1$

Q Use Runge-Kutta method of 4th order to evaluate $y(0.2)$ and $y(0.4)$ from the given i.v.p $\frac{dy}{dx} = e^x + y$, $y(0) = 0$.

Sol: Given: $\frac{dy}{dx} = e^x + y \Rightarrow f(x, y) = e^x + y$

$$y(0) = 0 \Rightarrow x_0 = 0, y_0 = 0.$$

We need to find y at 0.2 and y at 0.4 . So, we choose $h = 0.2$.

Then x -values are,

$$x_0 = 0, x_1 = 0.2, x_2 = 0.4.$$

Thus, we need to find y_1 and y_2 .

To find y_1 i.e. $y(0.2)$

Put $n = 0$ in ① & ②, then we get

$$y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad \text{--- ③}$$

$$k_1 = h f(x_0, y_0) = 0.2 f(0, 0) = 0.2 [e^0 + 0] = 0.2 \quad \boxed{f(x, y) = e^x + y}$$

$$k_2 = h f\left(x_0 + \frac{0.2}{2}, y_0 + \frac{0.2}{2}\right) = 0.2 f(0.1, 0.1) = 0.24103$$

$$k_3 = h f\left(x_0 + \frac{0.2}{2}, y_0 + \frac{0.24103}{2}\right) = 0.2 f(0.1, 0.1205) = 0.24514$$

$$k_4 = h f(x_0 + h, y_0 + k_3) = 0.2 f(0.2, 0.24514) = 0.29331$$

Substituting these values in ③,

$$y_1 = 0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$= 0.24428 \quad \text{i.e. } \underline{\underline{y(0.2) = 0.24428}}$$

R-K method of 4th order

$$y' = \frac{dy}{dx} = f(x, y), y(x_0) = y_0$$

$$y_{n+1} = y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad \text{--- ①}$$

$$\left. \begin{aligned} k_1 &= h f(x_n, y_n) \\ k_2 &= h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right) \\ k_3 &= h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right) \\ k_4 &= h f(x_n + h, y_n + k_3) \end{aligned} \right\} \quad \text{--- ②}$$

To find y_2 [ie $y(0.4)$]

Put $n=1$ in equ (1) & (2), then we get

$$y_2 = y_1 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4) \text{ --- (4) where}$$

$$K_1 = h f(x_1, y_1) = 0.2 f(0.2, 0.24428) = 0.29314$$

$$K_2 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{K_1}{2}\right) = 0.34814$$

$$K_3 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{K_2}{2}\right) = 0.35364$$

$$K_4 = h f(x_1 + h, y_1 + K_3) = 0.4179482$$

Substituting all values in equ. (4),

$$y_2 = 0.24428 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

$$= 0.59672$$

$$\text{ie } \underline{\underline{y(0.4) = 0.59672}}$$

Adams - Moulton Method: (Predictor-Correction method)

To solve o.d.e for getting solution y_{n+1} we proceed in two steps called predicting step and correcting step.

Given o.d.e is of the form :

$$y' = \frac{dy}{dx} = f(x, y) \quad ; \quad y(x_0) = y_0, y(x_1) = y_1, \dots, y(x_n) = y_n$$

To find y_{n+1} 1st we find approximate solution $y_{n+1}^{(P)}$ as follows:

Predicting step:

$$y_{n+1}^P = y_n + \frac{h}{24} [55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}]$$

Correcting step

In this step, we correct the predicted value y_{n+1}^P using the correcting formula,

$$y_{n+1} = y_n + \frac{h}{24} [9f_{n+1}^P + 19f_n - 5f_{n-1} + f_{n-2}]$$

where $f_{n+1}^P = f(x_{n+1}, y_{n+1}^P)$

(26)

①

9

Given: $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$,
 $y(1.2) = 1.548$, $y(1.3) = 1.979$

Find $y(1.4)$ using Adams-Moulton method.

Solution:

Adams-Moulton Method

$$\text{i.v.p : } \left. \begin{aligned} \frac{dy}{dx} &= f(x, y), & y(x_0) &= y_0 \\ & & y(x_1) &= y_1 \\ & & \vdots & \\ & & y(x_n) &= y_n \end{aligned} \right\} \text{--- (1)}$$

$$\text{predictor formula : } y_{n+1}^p = y_n + \frac{h}{24} [55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}] \text{--- (2)}$$

$$\text{corrector formula : } y_{n+1} = y_n + \frac{h}{24} [9f_{n+1}^p + 19f_n - 5f_{n-1} + f_{n-2}] \text{--- (3)}$$

$$f_{n+1} = f(x_{n+1}, y_{n+1}) \text{--- (4)}$$

Comparing given i.v.p with ①

$$f(x, y) = x^2(1+y)$$

$$y(1) = 1 \Rightarrow y_0 = 1$$

$$y(1.1) = 1.233 \Rightarrow y_1 = 1.233$$

$$y(1.2) = 1.548 \Rightarrow y_2 = 1.548$$

$$y(1.3) = 1.979 \Rightarrow y_3 = 1.979$$

$$y(1.4) = ? \text{ or } y_4 = ?$$

Note

From the given conditions, we see that values of x increases by 0.1
 i.e. step-size $h = 0.1$

$$\left\{ \begin{array}{ll} x_0 = 1 & x_3 = 1.3 \\ x_1 = 1.1 & x_4 = 1.4 \\ x_2 = 1.2 & \end{array} \right.$$

(27)

To find y_4 ie $y(1.4)$

Step ① (Predicting step)

In ② putting $n=3$,

$$y_4^P = y_3 + \frac{h}{24} [55f_3 - 59f_2 + 37f_1 - 9f_0] \quad \text{--- (5)}$$

To find f_0, f_1, f_2, f_3

$$f_0 = f(x_0, y_0) = f(1, 1) = 1^2(1+1) = 2 \quad \boxed{f(x, y) = x^2(1+y)}$$

$$f_1 = f(x_1, y_1) = f(1.1, 1.233) = (1.1)^2 [1 + 1.233] = 2.70193$$

$$f_2 = f(x_2, y_2) = f(1.2, 1.548) = (1.2)^2 [1 + 1.548] = 3.66912$$

$$f_3 = f(x_3, y_3) = f(1.3, 1.979) = (1.3)^2 [1 + 1.979] = 5.03451$$

Substituting all these values in (5),

$$y_4^P = 1.979 + \frac{0.1}{24} [55 \times 5.03451 - 59 \times 3.66912 + 37 \times 2.70193 - 9 \times 2]$$

$$y_4^P = \underline{2.572297}$$

~~Step~~ Step ② (Correcting step)

In ③, put $n=3$,

$$y_4 = y_3 + \frac{h}{24} [9f_4^P + 19f_3 - 5f_2 + f_1]$$
$$= 2.5749473$$

ie $y(1.4) = \underline{\underline{2.5749473}}$

$$\left\{ \begin{aligned} f_4^P &= f(x_4, y_4^P) \\ &= f(1.4, 2.572297) \\ &= (1.4)^2 [1 + 2.572297] \\ &= 7.0017029 \end{aligned} \right.$$

- Q Apply Adams Moulton method to the initial value problem $y' = x + y$, $y(0) = 0$, $y(0.2) = 0.0214$, $y(0.4) = 0.91818$, $y(0.6) = 0.222107$ to obtain the solutions at $x = 0.8$ and $x = 1$.

Solution

Adams-Moulton Method

$$\text{i.v.p : } \left. \begin{array}{l} \frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \\ y(x_1) = y_1 \\ \vdots \\ y(x_n) = y_n \end{array} \right\} \rightarrow (1)$$

$$(x_n = x_0 + nh)$$

$$\text{predictor formula} : y_{n+1}^p = y_n + \frac{h}{24} [55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}] \rightarrow (2)$$

$$\text{corrector formula} : y_{n+1} = y_n + \frac{h}{24} [9f_{n+1}^p + 19f_n - 5f_{n-1} + f_{n-2}] \rightarrow (3)$$

$$f_{n+1} = f(x_{n+1}, y_{n+1}) \rightarrow (4)$$

Comparing given i.v.p with (1),

$$f(x, y) = x + y$$

$$y(0) = 0 \Rightarrow y_0 = 0$$

$$y(0.2) = 0.0214 \Rightarrow y_1 = 0.0214$$

$$y(0.4) = 0.91818 \Rightarrow y_2 = 0.91818$$

$$y(0.6) = 0.222107 \Rightarrow y_3 = 0.222107$$

$$y(0.8) = ? \quad \text{ie } y_4 = ?$$

$$y(1) = ? \quad \text{ie } y_5 = ?$$

Note:

From the given conditions we see that value of x increases by 0.2 ie step size $h = 0.2$

$$x_0 = 0$$

$$x_1 = 0.2$$

$$x_2 = 0.4$$

$$x_3 = 0.6$$

$$x_4 = 0.8$$

$$x_5 = 1$$

To find y_4 ie $y(0.8)$

(29)

Step ① Predicting step

In ②, putting $n=3$,

$$y_4^P = y_3 + \frac{h}{24} [55f_3 - 59f_2 + 37f_1 - 9f_0] \quad \text{--- (5)}$$

To find f_0, f_1, f_2, f_3

$$f_0 = f(x_0, y_0) = f(0, 0) = 0 + 0 = 0$$

$$\boxed{f(x, y) = x + y}$$

$$f_1 = f(x_1, y_1) = f(0.2, 0.0214) = 0.2 + 0.0214 = 0.2214$$

$$f_2 = f(x_2, y_2) = f(0.4, 0.091818) = 0.4 + 0.091818 = 0.491818$$

$$f_3 = f(x_3, y_3) = f(0.6, 0.222107) = 0.6 + 0.222107 = 0.822107$$

Substituting all these values in (5),

$$y_4^P = 0.222107 + \frac{0.2}{24} [55f_3 - 59f_2 + 37f_1 - 9f_0] = 0.425361$$

$$\text{ie } y_4^P = \underline{\underline{0.425361}}$$

Step ② Correcting step

In ③ putting $n=3$,

$$y_4 = y_3 + \frac{h}{24} [9f_4^P + 19f_3 - 5f_2 + f_1]$$

$$= 0.222107 + \frac{0.2}{24} [9 \times 1.225361 + 19f_3 - 5f_2 + f_1]$$

$$= 0.425529$$

$$\text{ie } y(0.8) = \underline{\underline{0.425529}} = y_4$$

ie solution of ode
at $x=0.8$ is
 $0.425529 //$

$$f(x, y) = x + y$$

$$f_4^P = f(x_4, y_4^P)$$

$$= x_4 + y_4^P$$

$$= 0.8 + 0.425361$$

$$= 1.225361$$

(30) (5)

To find y_5 ie $y(1)$

Step ① Predicting step

In ② putting $n=4$

$$y_5^P = y_4 + \frac{h}{24} [55f_4 - 59f_3 + 37f_2 - 9f_1] \quad \text{--- ⑥}$$

we already have values of f_1, f_2, f_3 .

To find f_4 ; $f_4 = f(x_4, y_4)$

$$= f(0.8, 0.425529)$$

$$= 0.8 + 0.425529 = 1.225529$$

$$\boxed{f(x, y) = x + y}$$

Substituting all values in ⑥,

$$y_5^P = 0.425529 + \frac{0.2}{24} [55 \times 1.225529 - 59f_3 + 37f_2 - 9f_1]$$
$$= \underline{\underline{0.718066}}$$

Step ② Correcting step

In ③, put $n=4$,

$$y_5 = y_4 + \frac{h}{24} [9f_5^P + 19f_4 - 5f_3 + f_2]$$

$$= 0.4255286 + \frac{0.2}{24} [9 \times 1.718066 + 19f_4 - 5f_3 + f_2]$$

$$= 0.718270$$

$$\left\{ \begin{array}{l} f(x, y) = x + y \\ f_5^P = f(x_5, y_5^P) \\ = x_5 + y_5^P \\ = 1 + 0.718066 \\ = 1.718066 \end{array} \right.$$

ie $y(1) = 0.718270$

ie value of y or solution at $x=1$ is $\underline{\underline{y_5 = 0.718270}}$

Q4 Using Adams-Moulton method, find $y(0.8)$ from the initial value problem $\frac{dy}{dx} = 1+y^2$, $y(0)=0$, $y(0.2)=0.2027$, $y(0.4)=0.4228$, $y(0.6)=0.6841$

Solution

Adams-Moulton Method

$$\text{i.v.p : } \left. \begin{aligned} \frac{dy}{dx} &= f(x, y), \quad y(x_0) = y_0 \\ y(x_1) &= y_1 \\ y(x_n) &= y_n \end{aligned} \right\} \rightarrow (1)$$

$$\text{prediction formula : } y_{n+1}^p = y_n + \frac{h}{24} [45f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}] \rightarrow (2)$$

$$\text{correction formula : } y_{n+1} = y_n + \frac{h}{24} [9f_{n+1}^p + 19f_n - 5f_{n-1} + f_{n-2}] \rightarrow (3)$$

$$f_{n+1} = f(x_{n+1}, y_{n+1}) \rightarrow (4)$$

Comparing given i.v.p with (1),

$$f(x, y) = 1+y^2$$

$$y(0)=0 \Rightarrow y_0=0$$

$$y(0.2)=0.2027 \Rightarrow y_1=0.2027$$

$$y(0.4)=0.4228 \Rightarrow y_2=0.4228$$

$$y(0.6)=0.6841 \Rightarrow y_3=0.6841$$

$$y(0.8) = ? \Rightarrow y_4 = ?$$

We have to find $y(0.8)$. i.e. we've to find y_4 .

To find y_4

Step (1) (prediction step)

First we'll first predict the value of y_4 using the predictor formula (2)

In (2), putting $n=3$, we get y_4^p which is the predictor value of y_4 .

Note:
From the given conditions, we see that value of 'x' increases by 0.2 i.e. step size $h=0.2$

$$y_4^P = y_3 + \frac{h}{24} [55f_3 - 59f_2 + 37f_1 - 9f_0] \quad \text{--- (5)}$$

To find f_0, f_1, f_2, f_3

$$f_0 = f(x_0, y_0) = f(0, 0) = 1 + 0^2 = \underline{\underline{1}}$$

$$f_1 = f(x_1, y_1) = f(0.2, 0.2027) = 1 + (0.2027)^2 = 1.0411$$

$$f_2 = f(x_2, y_2) = f(0.4, 0.4228) = 1 + (0.4228)^2 = 1.1788$$

$$f_3 = f(x_3, y_3) = f(0.6, 0.6841) = 1 + (0.6841)^2 = 1.4679$$

Substituting all these values in (5),

$$y_4^P = 0.6841 + \frac{0.2}{24} [55 \times 1.4679 - 59 \times 1.1788 + 37 \times 1.0411 - 9 \times 1]$$

$$= 0.6841 + \frac{0.2}{24} [80.7345 - 69.5492 + 38.5207 - 9]$$

$$= \underline{\underline{1.0233}}$$

Step 2 Correcting step

In (3), put $n=3$,

$$y_4 = y_3 + \frac{h}{24} [9f_4^P + 19f_3 - 5f_2 + f_1]$$

$$= 0.6841 + \frac{0.2}{24} [9 \times 2.04714 + 19 \times 1.4679 - 5 \times 1.1788 + 1.0411]$$

$$= 0.6841 + \frac{0.2}{24} [18.42426 + 27.8901 - 5.894 + 1.0411]$$

$$= \underline{\underline{1.0296}} \quad \text{ie } y(0.8) = \underline{\underline{1.0296}}$$

we've (given)	
$x_0 = 0$	$y_0 = 0$
$x_1 = 0.2$	$y_1 = 0.2027$
$x_2 = 0.4$	$y_2 = 0.4228$
$x_3 = 0.6$	$y_3 = 0.6841$
$x_4 = 0.8$	$y_4 = ?$

$$f_4^P = f(x_4, y_4^P)$$

$$f(x, y) = 1 + y^2$$

$$\therefore f_4^P = 1 + (y_4^P)^2$$

$$= 1 + (1.0233)^2$$

$$= 2.04714$$